

Task 1 -Foundation & Environment Setup

Introduction:

- **Internship:** Cyber Security Internship at ApexPlanet
- **Task:** Foundation & Environment Setup
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Objective:

The objective of this task is to build a basic cybersecurity laboratory environment using virtualization technologies. The lab consists of Kali Linux as the attacker machine and Metasploitable2 as the vulnerable target machine. The setup helps in understanding networking concepts, Linux fundamentals, cybersecurity basics, and security assessment tools such as Nmap and Wireshark.

Cybersecurity Basics:

1. CIA Traid:

- **Confidentiality** : Ensures information is accessible only to authorized users
- **Integrity** : Ensures information is not modified by unauthorized users
- **Availability** : Ensures the information and services are available whenever needed.

Threats :

- **Phishing** : A social engineering attack to trick user to get their credentials.
- **Malware** : Malicious software designed to gain unauthorized access to systems
- **DDoS**: Distributed Denial of Service to make a server can't handle the traffic
- **SQL Injection** : Attack related database queries to get sensitive data from it

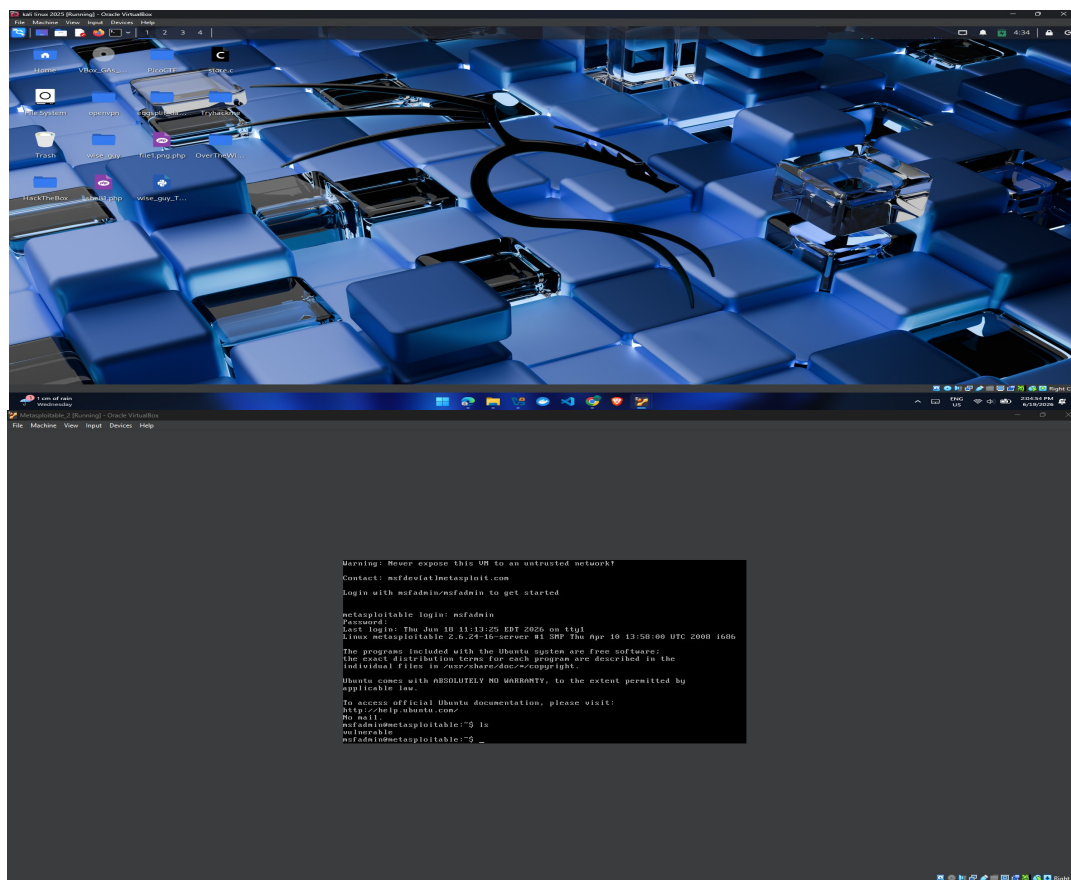
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- **Brute Force** : Guessing password until one is right with software or programs
- **Ransomware** : Malware that encrypts files and demands payment for decryption.

2. Lab Environment Setup :

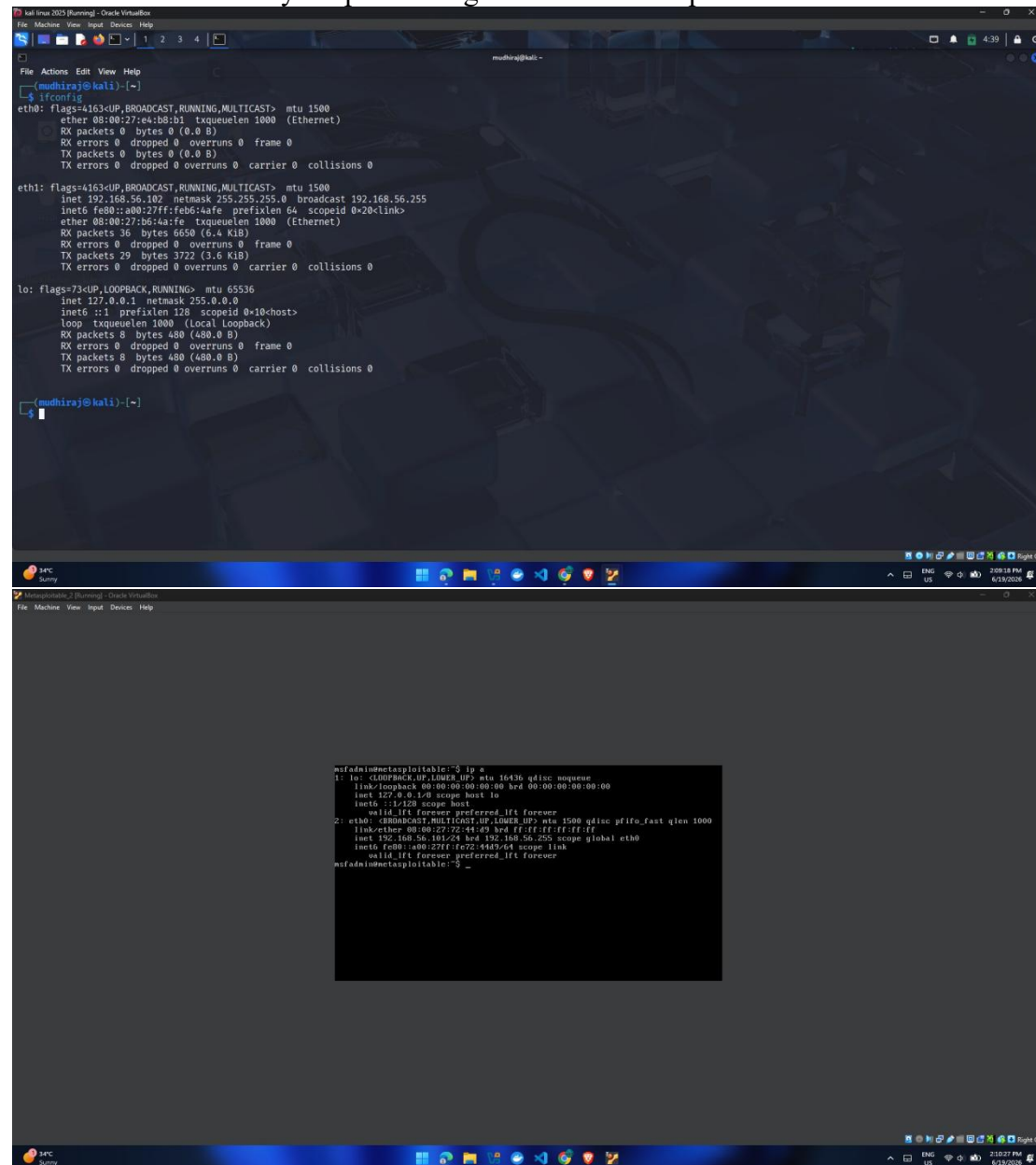
- **Software Used :**
 - ◆ “Oracle VritualBox”
 - ◆ “Kali Linux 2025”
 - ◆ “Metasploitable”

They can be found on the screen shots below.



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Below are the host only adapter setting in kali and metasploitable



```
(mudhiraj@kali)~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    ether 08:00:27:e4:1b:81 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.102 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::a00:27ff:feb6:4afe prefixlen 64 scopeid 0<20<link>
    ether 08:00:27:b6:4a:fe txqueuelen 1000 (Ethernet)
    RX packets 36 bytes 6650 (6.4 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 29 bytes 3722 (3.6 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(mudhiraj@kali)~$

msfadmin@metasploitable:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        inet6 ::1/128 scope host
            valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 08:00:27:72:44:49 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.101/24 brd 192.168.56.255 scope global eth0
        inet6 fe80::a00:27ff:fe72:4449/64 scope link
            valid_lft forever preferred_lft forever
msfadmin@metasploitable:~$
```

eth1 and eth0 in kali and metasploitable are in host only adapter they are in same network with VM DHCP sever.

Kali Linux had two adapter 1 and 2 with NAT and Host-Only one .
Metasploitable 2 has one adapter with Host-Only adapter only.

NAT Adapter : NAT (Network Address Translation) allows private IP addresses to access the internet using a public IP address.

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Host-Only Adapter : Host-Only networking creates a private network between virtual machines.

3. Networking Basics :

OSI LAYER

Layer	Functions	Examples
Application layer	UI/UX	chrome
Presentation Layer	Encryption/formatting/compression	SSL/TLS
Session layer	Maintains connecyion	Login session
Transport layer	Reliable delivery	TCP,UDP
Network Layer	Routing using ip addr	IP address
DataLink layer	MAC addr communication	Etherent
Physical Layer	Actual signals	Cable/Wifi

TCP/IP Protocol Suite

Same as OSI but has 4 layer with integrated into them

TCP/IP	Equals to OSI layer
Application layer	7,6,5
Transport layer	4
Internet layer	3
Network layer	2,1

DNS Deep Dive :

DNS converts domain names into IP addresses so computers can locate websites.

HTTP/HTTPS Deep Dive:

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- * Hyper Text Transfer Protocol is webpages transfer protocol used in web browsers.
- * In HTTPS it will transfer with encryption, so no middle man will see the text you sent over the network. Here 's' indicates secure travel in internet.

IP Addressing :

- * It is like a address to specifly indentifies a system in network, In real life it is home address.

Subnetting :

Subnetting is the process of dividing a large network into smaller logical networks for better management and communication.

NAT :

- * Network Address Translation convert public IP to private IP for internet browsing.

My lab Example :

Kali Linux :

eth0 = NAT

Internet Access

eth1 = Host-Only

192.168.56.102

Metasploitable2 :

eth0 = Host-Only

192.168.56.101

Communication is happening in eth1 and eth0 of kali and Metasploitable 2.

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4 .Cryptography Basics

Cryptography – Technique used to secure data from unauthorized access.

Symmetric Encryption – Uses the same key for encryption and decryption.

Asymmetric Encryption – Uses a public key for encryption and a private key for decryption.

AES – A popular symmetric encryption algorithm.

RSA – A popular asymmetric encryption algorithm.

Hashing – Converts data into a fixed-length value that cannot be reversed.

MD5 – A 128-bit hashing algorithm used for integrity checks.

SHA256 – A 256-bit secure hashing algorithm.

Digital Certificate – Electronic document used to verify the identity of a website or organization.

Certificate Authority (CA) – Trusted organization that issues digital certificates.

SSL – Older protocol used to secure network communication.

TLS – Modern protocol that secures data transmission over networks.

HTTPS – HTTP secured using SSL/TLS encryption.

OpenSSL – Open-source tool used for encryption, decryption, and certificate management.

Hands-on OpenSSL :

Encryption Command :

```
openssl enc -aes-256-cbc -in message.txt -out encrypted.enc
```

Encrypts a file using AES-256 encryption.

Decryption Command :

```
openssl enc -aes-256-cbc -d -in encrypted.enc -out decrypted.txt
```

Decrypts the encrypted file back to its original form.

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Conclusion:s

Successfully installed VirtualBox, Kali Linux, and Metasploitable2. Configured NAT and Host-Only networking, verified connectivity using ping, identified services using Nmap, and captured network packets using Wireshark. This lab environment will be used for future cybersecurity learning and penetration testing exercises.